

Master Thesis
BlueRov2 Formation Control

Week 2

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Previous Week

- Formation control with single integrator agents no constraints
  - Undirected graph
  - Directed graph
- Single integrator formation control with constraints
  - Only connectivity maintenance
  - Then also colision avoidance
- Double integrator agents
- BlueRov2 model no input constraints
- [Research challenge] integrate input constraints
- [Research challenge] controller robust or less sensitive to uncertainties (e.g. inverse optimal controller in the sense of LgV like control laws rather than feedback linearization)

Achievements

- Study
  - Graph Theoretic Methods in Multiagent Networks
  - Esteban’s thesis

- Implementation
  - Improvements
    - Organized the code into different files
    - Generalized the code
  - Added Constraints
    - Connectivity Maintenance
    - Collision Avoidance

Implementation - Improvements

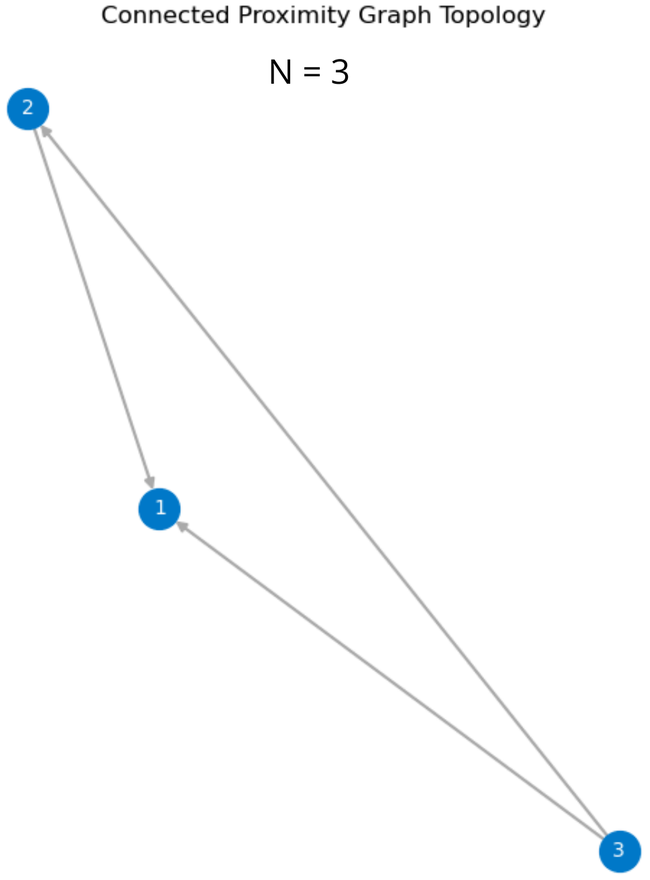
parameters.py - where parameteres are defined

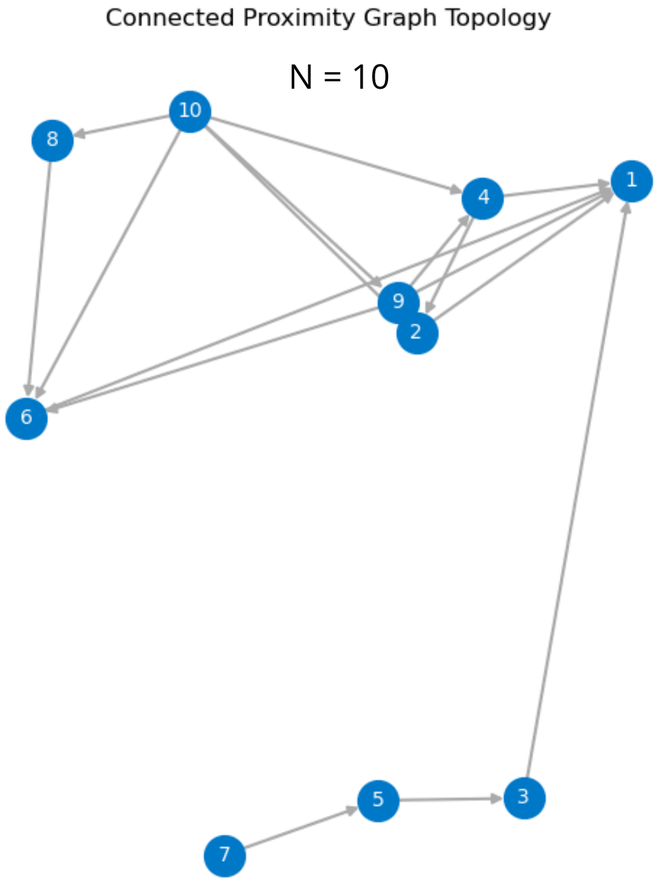
formation_controller.py - where all the computation regarding the agent’s formation is handled

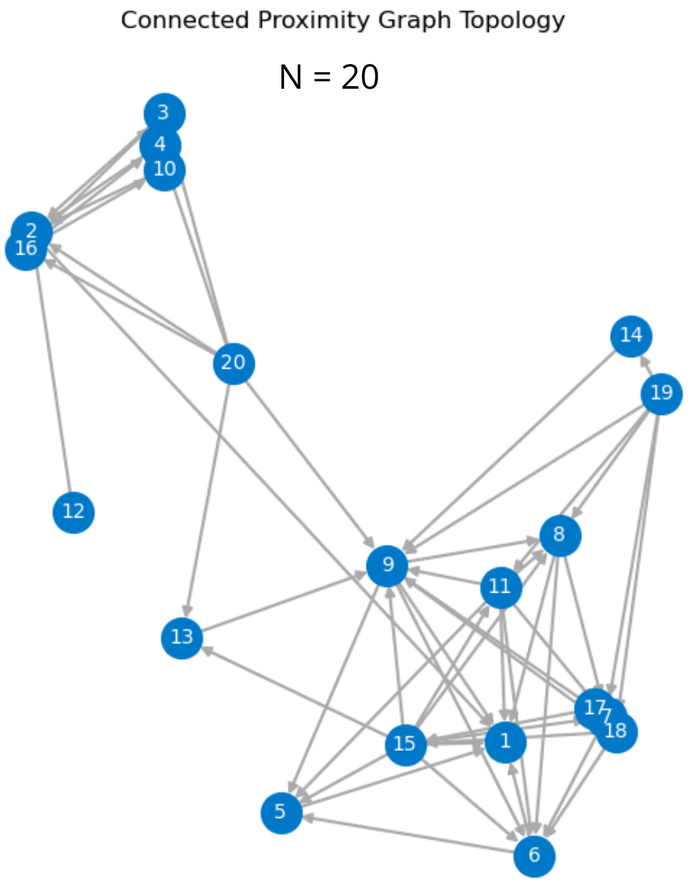
plot_animator.py - where the plots and animations are set up

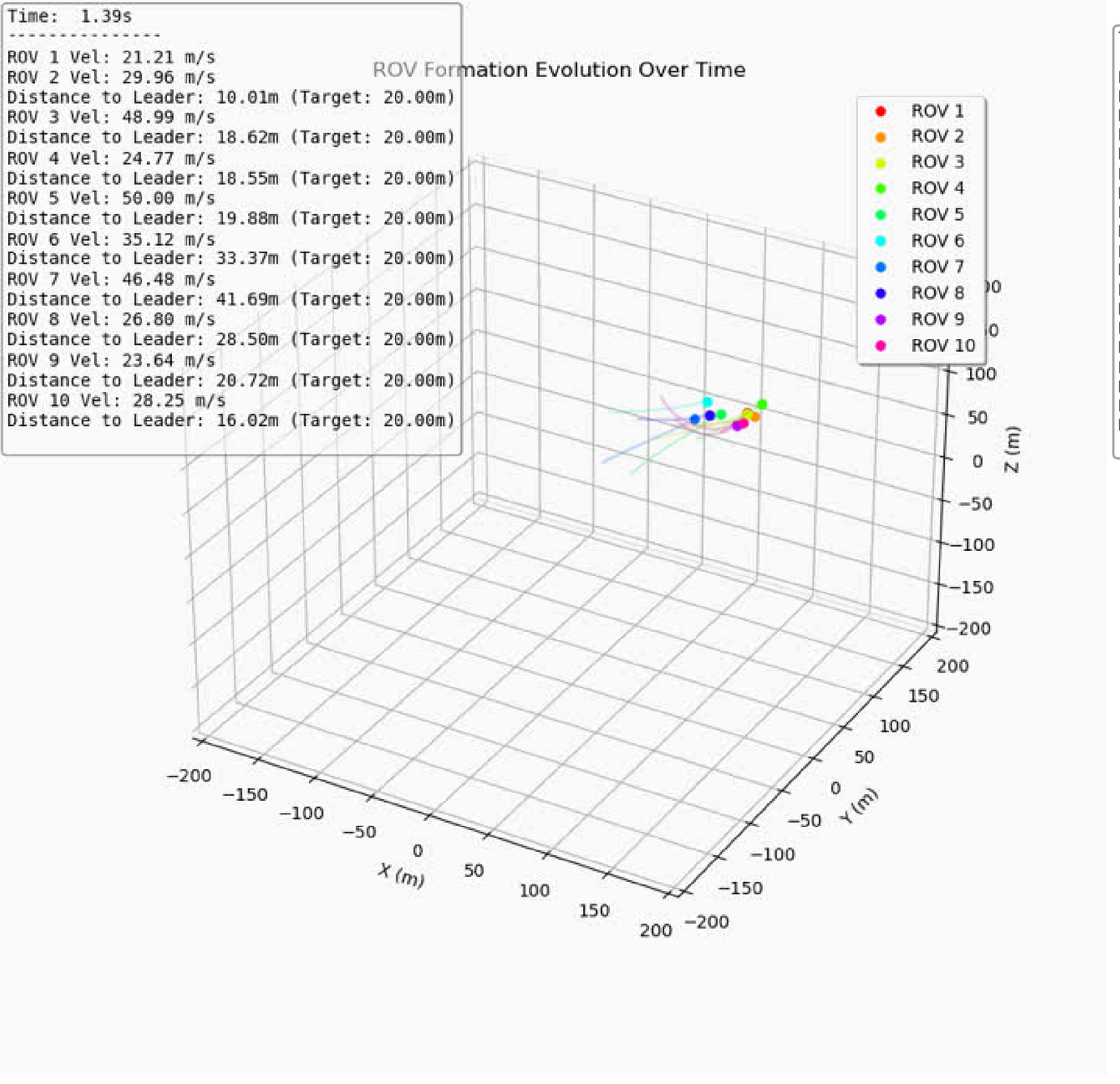
graphs_utils.py - where agent’s pose, graph and variables that depend on it are generated/computed

single_integrator_direct.py - where everything comes together (main)

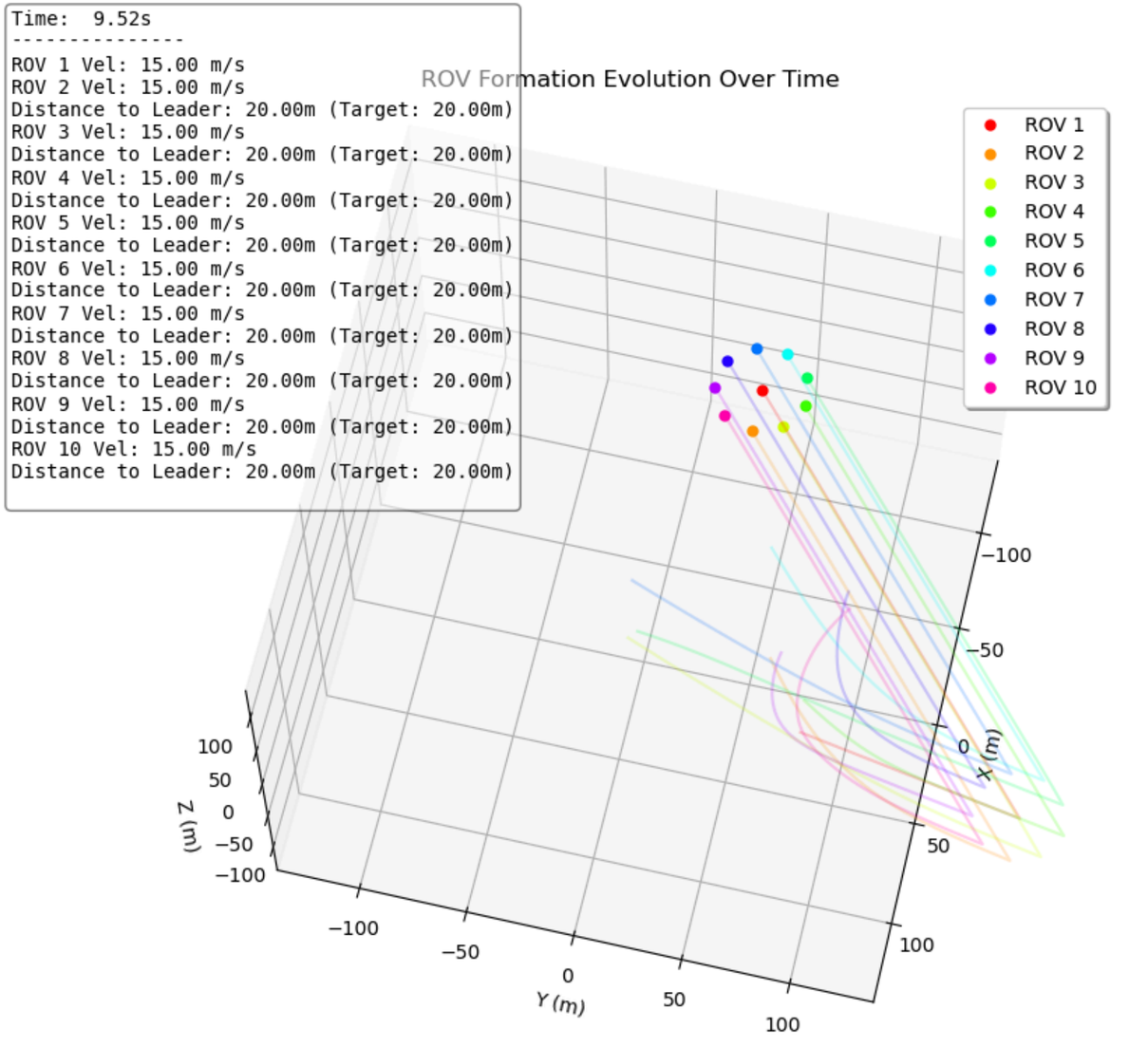

Connected Proximity Graph Topology
N = 3
2
1
3


Connected Proximity Graph Topology
N = 10
10
8
4
1
9
2
6
5
3
7


Connected Proximity Graph Topology
N = 20
3
4
10
2
16
14
20
19
12
8
9
11
13
17
7
18
15
1
5
6

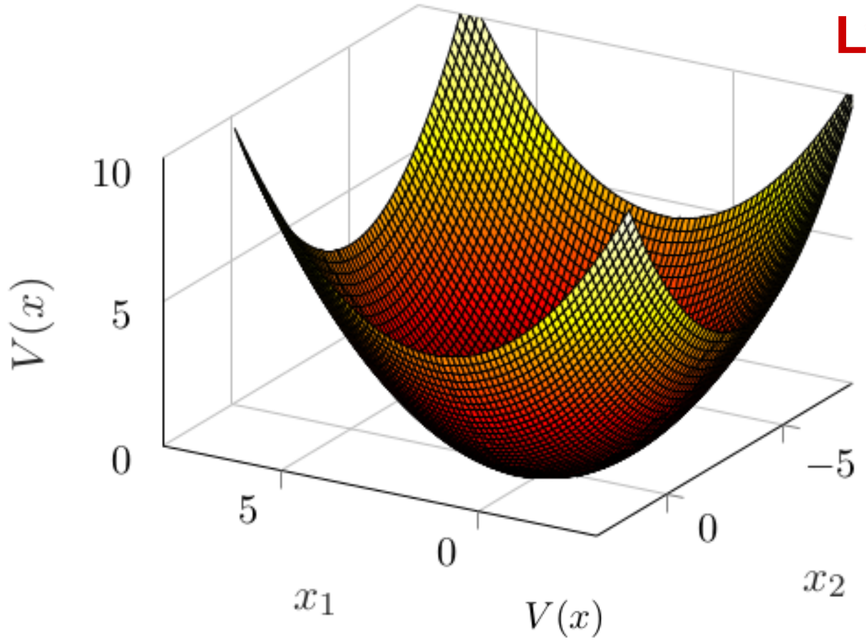

Time: 1.39s

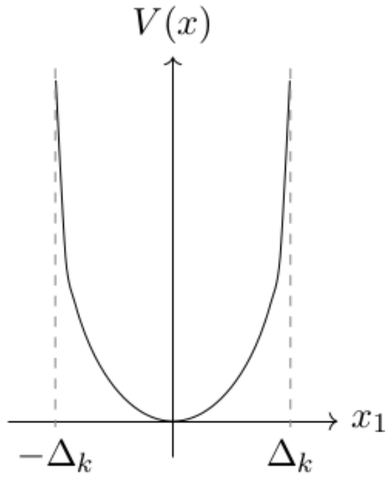
ROV 1 Vel: 21.21 m/s
ROV 2 Vel: 29.96 m/s
Distance to Leader: 10.01m (Target: 20.00m)
ROV 3 Vel: 48.99 m/s
Distance to Leader: 18.62m (Target: 20.00m)
ROV 4 Vel: 24.77 m/s
Distance to Leader: 18.55m (Target: 20.00m)
ROV 5 Vel: 50.00 m/s
Distance to Leader: 19.88m (Target: 20.00m)
ROV 6 Vel: 35.12 m/s
Distance to Leader: 33.37m (Target: 20.00m)
ROV 7 Vel: 46.48 m/s
Distance to Leader: 41.69m (Target: 20.00m)
ROV 8 Vel: 26.80 m/s
Distance to Leader: 28.50m (Target: 20.00m)
ROV 9 Vel: 23.64 m/s
Distance to Leader: 20.72m (Target: 20.00m)
ROV 10 Vel: 28.25 m/s
Distance to Leader: 16.02m (Target: 20.00m)
ROV Formation Evolution Over Time
ROV 1
ROV 2
ROV 3
ROV 4
ROV 5
ROV 6
ROV 7
ROV 8
ROV 9
ROV 10
X (m)
Y (m)
Z (m)


Time: 9.52s

ROV 1 Vel: 15.00 m/s
ROV 2 Vel: 15.00 m/s
Distance to Leader: 20.00m (Target: 20.00m)
ROV 3 Vel: 15.00 m/s
Distance to Leader: 20.00m (Target: 20.00m)
ROV 4 Vel: 15.00 m/s
Distance to Leader: 20.00m (Target: 20.00m)
ROV 5 Vel: 15.00 m/s
Distance to Leader: 20.00m (Target: 20.00m)
ROV 6 Vel: 15.00 m/s
Distance to Leader: 20.00m (Target: 20.00m)
ROV 7 Vel: 15.00 m/s
Distance to Leader: 20.00m (Target: 20.00m)
ROV 8 Vel: 15.00 m/s
Distance to Leader: 20.00m (Target: 20.00m)
ROV 9 Vel: 15.00 m/s
Distance to Leader: 20.00m (Target: 20.00m)
ROV 10 Vel: 15.00 m/s
Distance to Leader: 20.00m (Target: 20.00m)
ROV Formation Evolution Over Time
ROV 1
ROV 2
ROV 3
ROV 4
ROV 5
ROV 6
ROV 7
ROV 8
ROV 9
ROV 10
Z (m)
Y (m)
X (m)

Implementation - Connectivity Maintenance





Lyapunov Barrier Functions

$$v_i^* = -c_1 \sum_{k \leq M} [E_\odot]_{ik} \nabla W_k(z_k)$$

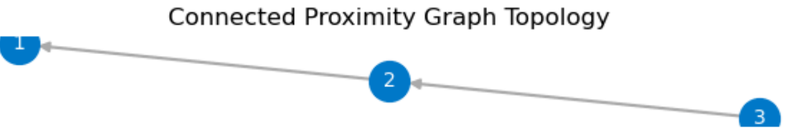
$$u_i = -c_2 \tilde{v}_i + \dot{v}_i^*,$$

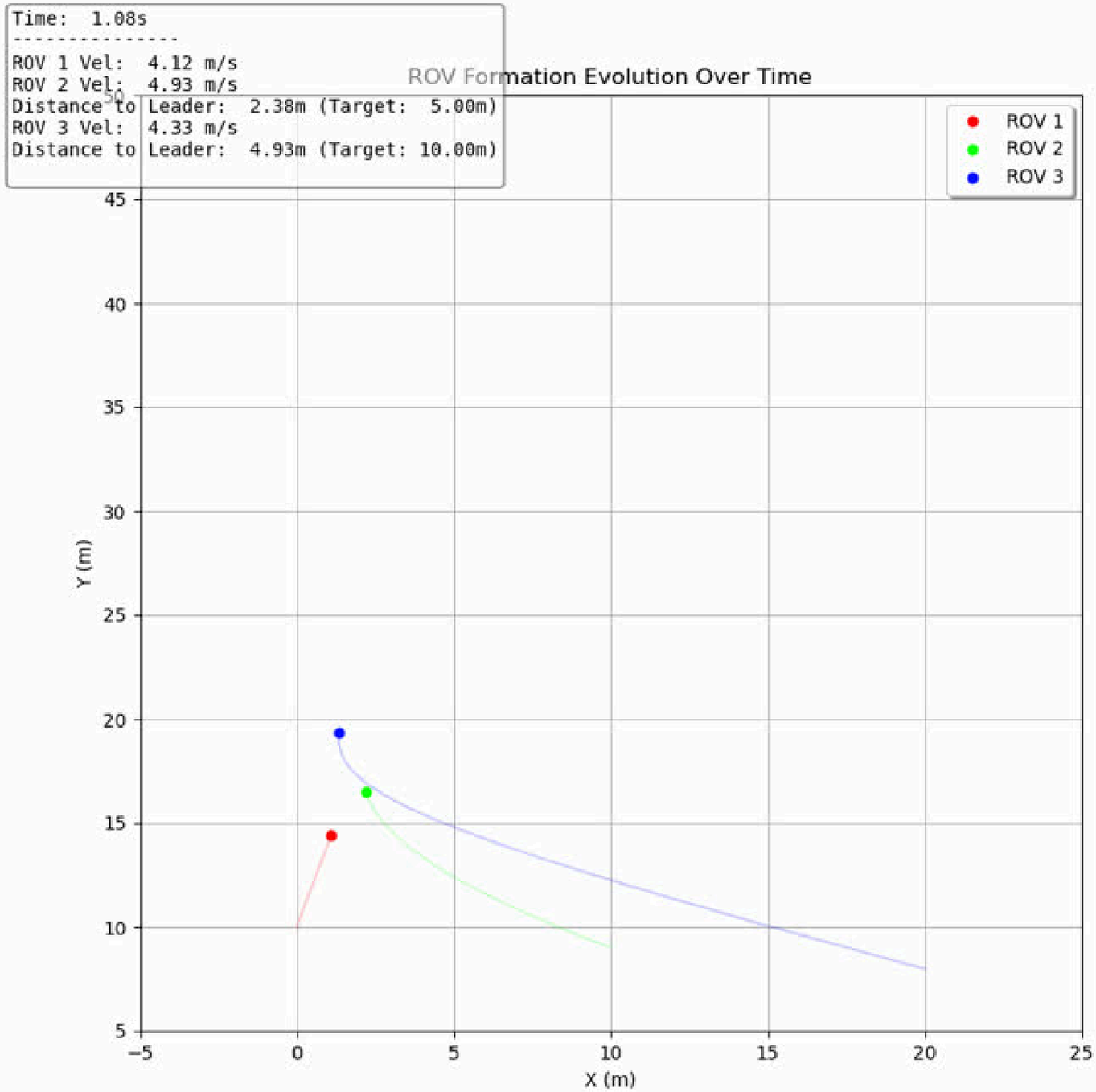
$$z_k := x_i - x_j$$

$$\nabla W_k(z_k) = \left[1 + \frac{1}{\Delta_k^2 - |z_k|^2}\right] z_k,$$

$$\tilde{v}_i = \sum_{j \in \mathcal{N}_i} [(x_i - x_j) - (o_i - o_j)]$$

Issues

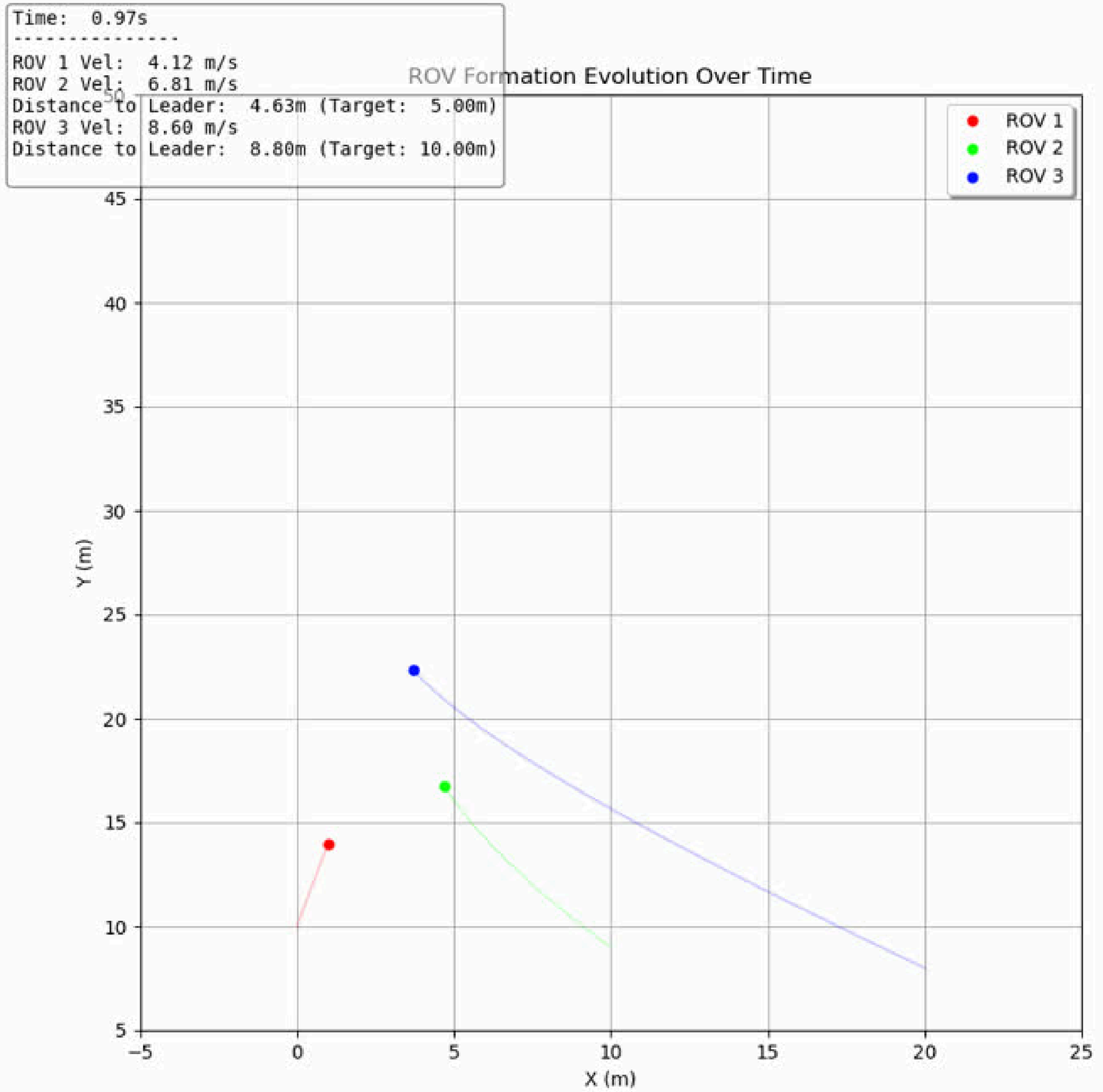

Connected Proximity Graph Topology
1
2
3


Time: 1.08s
ROV 1 Vel: 4.12 m/s
ROV 2 Vel: 4.93 m/s
Distance to Leader: 2.38m (Target: 5.00m)
ROV 3 Vel: 4.33 m/s
Distance to Leader: 4.93m (Target: 10.00m)
ROV Formation Evolution Over Time
ROV 1
ROV 2
ROV 3
Y (m)
X (m)
45
40
35
30
25
20
15
10
5
−5
0
5
10
15
20
25

Solution

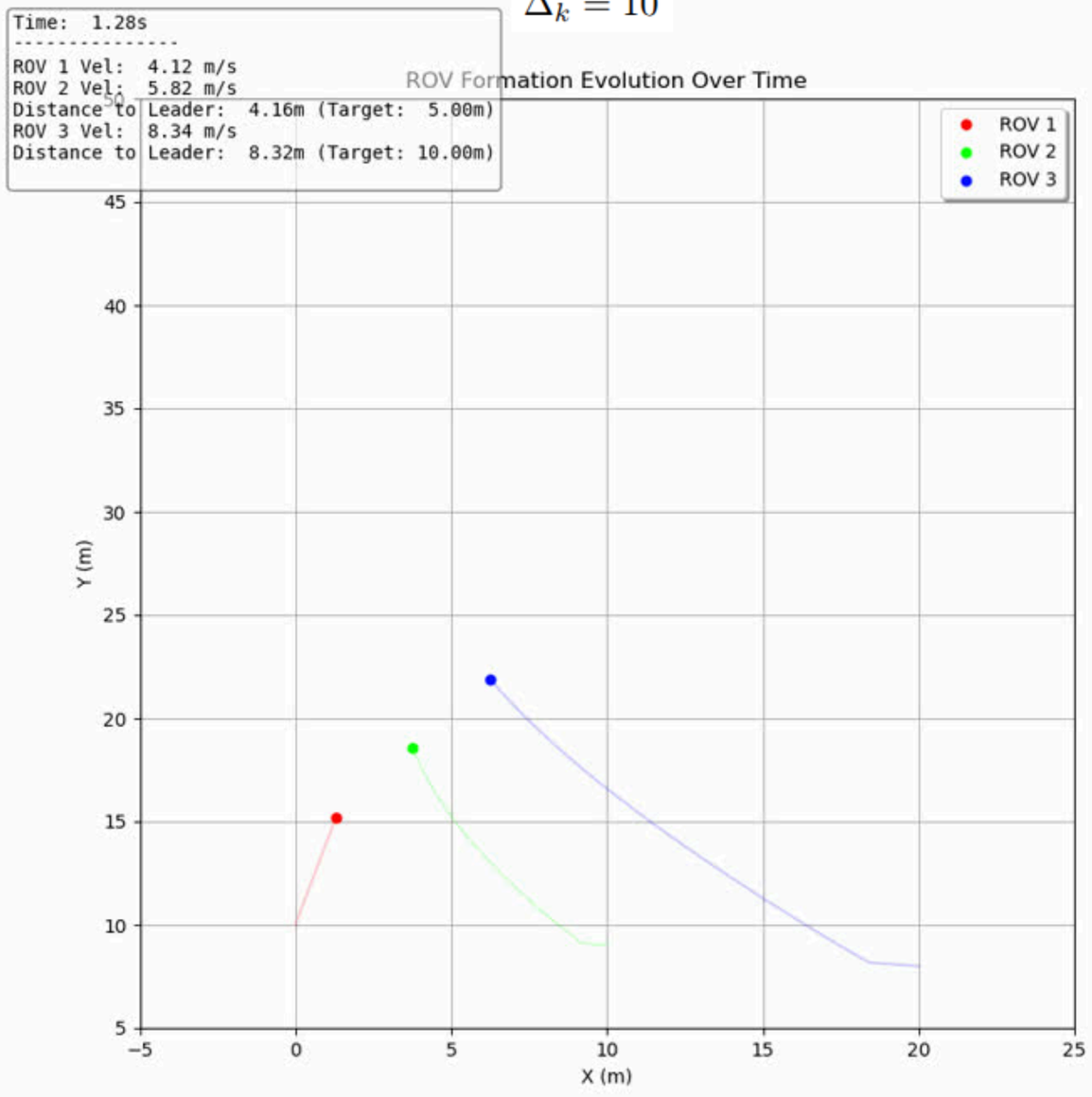
$$\nabla W_k(z_k) = \left[1 + \frac{1}{\Delta_k^2 - |z_k|^2}\right] z_k,$$

$$\nabla W_k(z_k) = \left(\frac{1}{\Delta_k^2 - |z_k|^2}\right) z_k$$

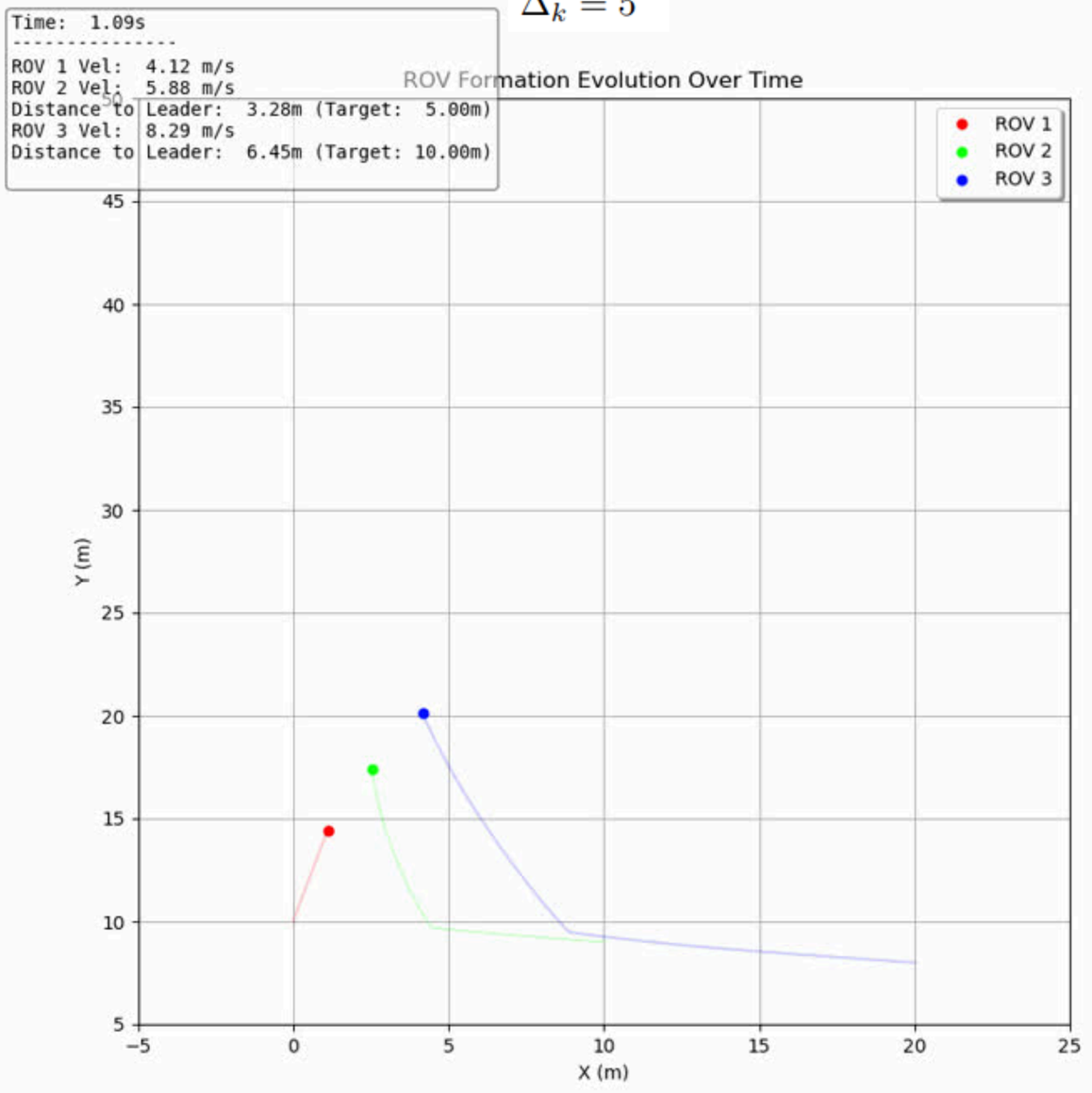

Time: 0.97s

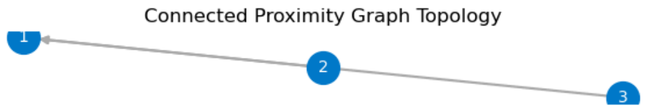
ROV 1 Vel: 4.12 m/s
ROV 2 Vel: 6.81 m/s
Distance to Leader: 4.63m (Target: 5.00m)
ROV 3 Vel: 8.60 m/s
Distance to Leader: 8.80m (Target: 10.00m)
ROV Formation Evolution Over Time
ROV 1
ROV 2
ROV 3
Y (m)
X (m)
50
45
40
35
30
25
20
15
10
5
−5
0
5
10
15
20
25

$\Delta_k = 10$

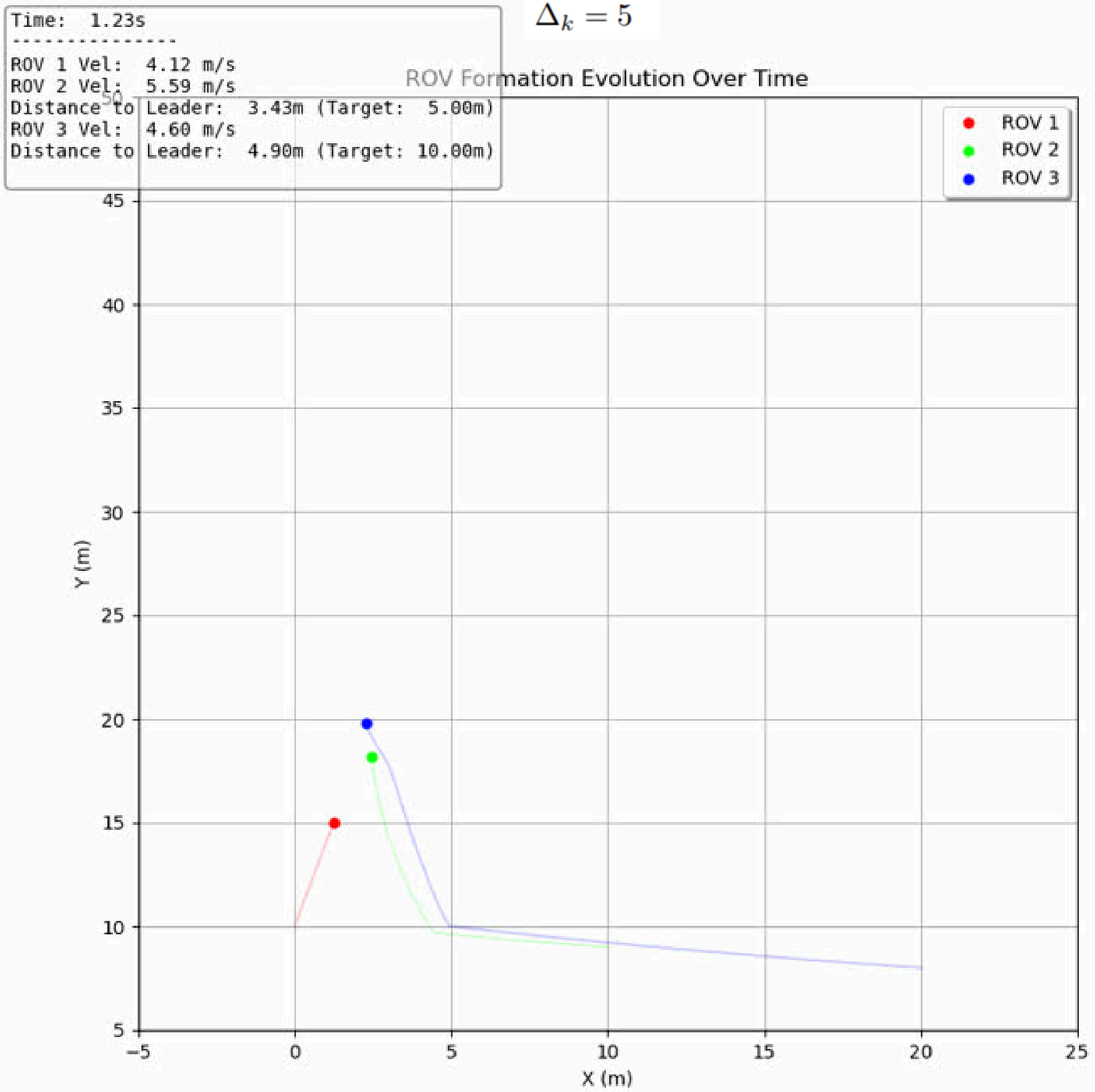


$\Delta_k = 5$



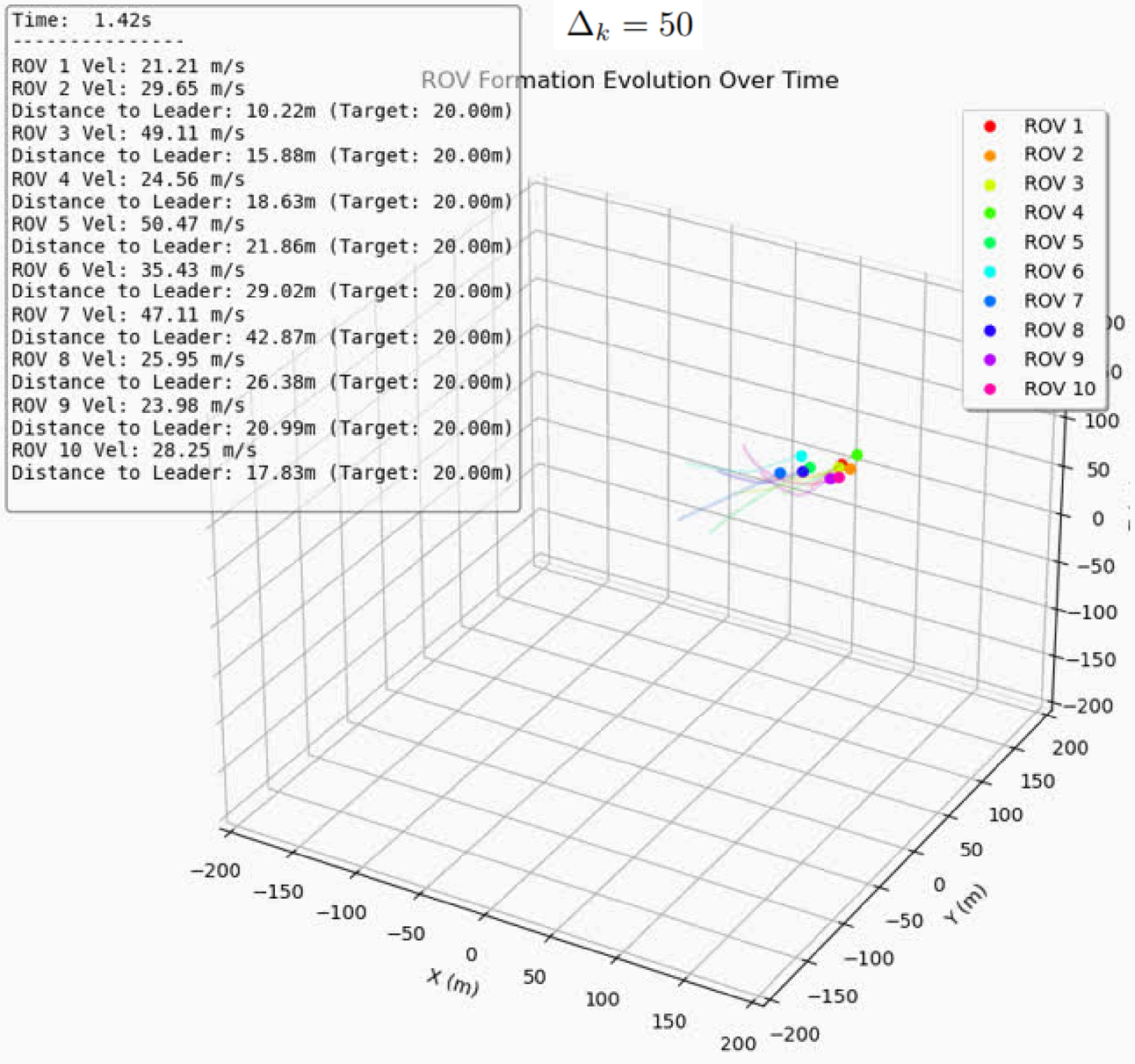

Connected Proximity Graph Topology
1
2
3

$\Delta_k = 5$

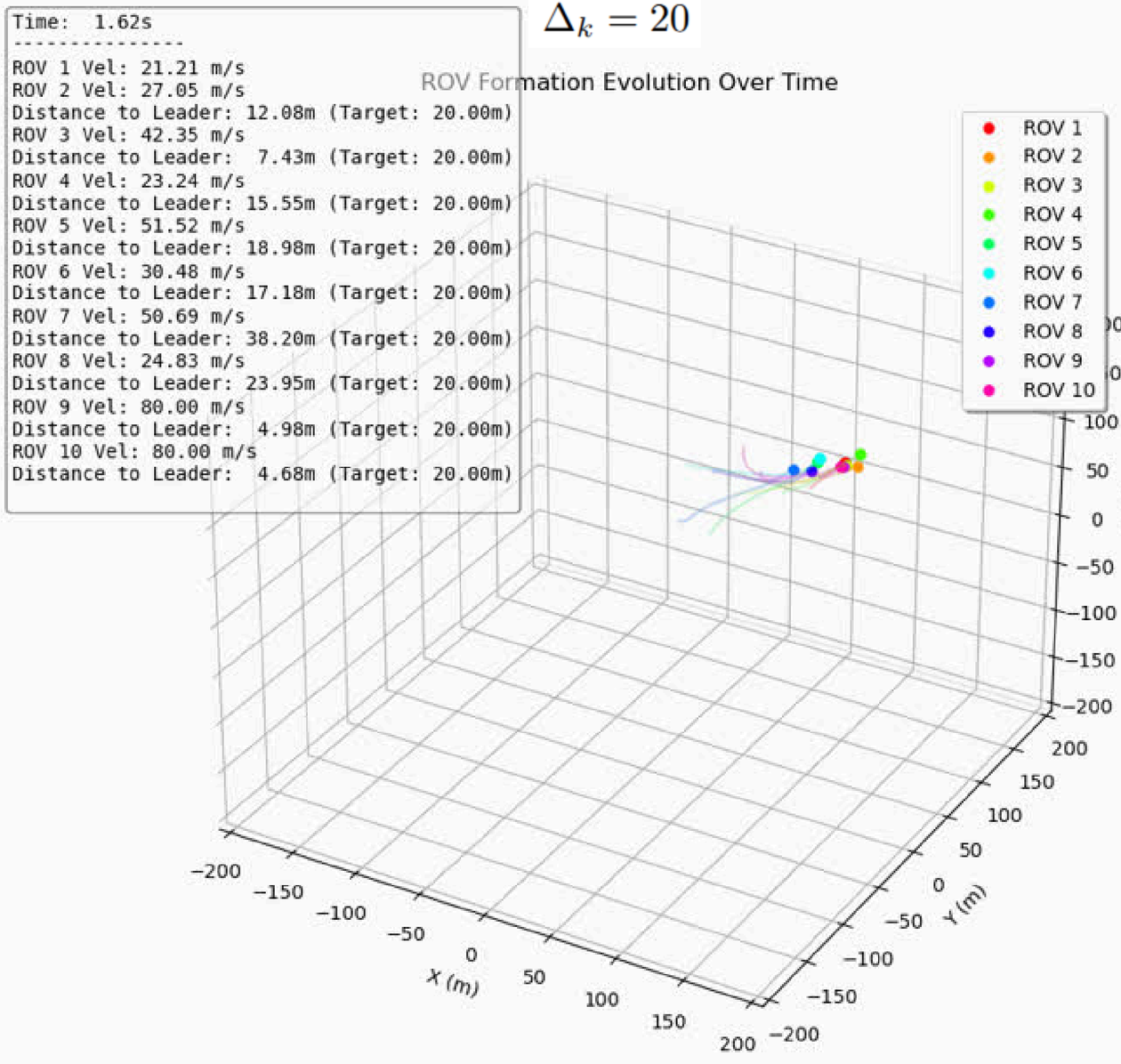

Time: 1.23s

ROV 1 Vel: 4.12 m/s
ROV 2 Vel: 5.59 m/s
Distance to Leader: 3.43m (Target: 5.00m)
ROV 3 Vel: 4.60 m/s
Distance to Leader: 4.90m (Target: 10.00m)
ROV Formation Evolution Over Time
ROV 1
ROV 2
ROV 3
Y (m)
X (m)
50
45
40
35
30
25
20
15
10
5
−5
0
5
10
15
20
25

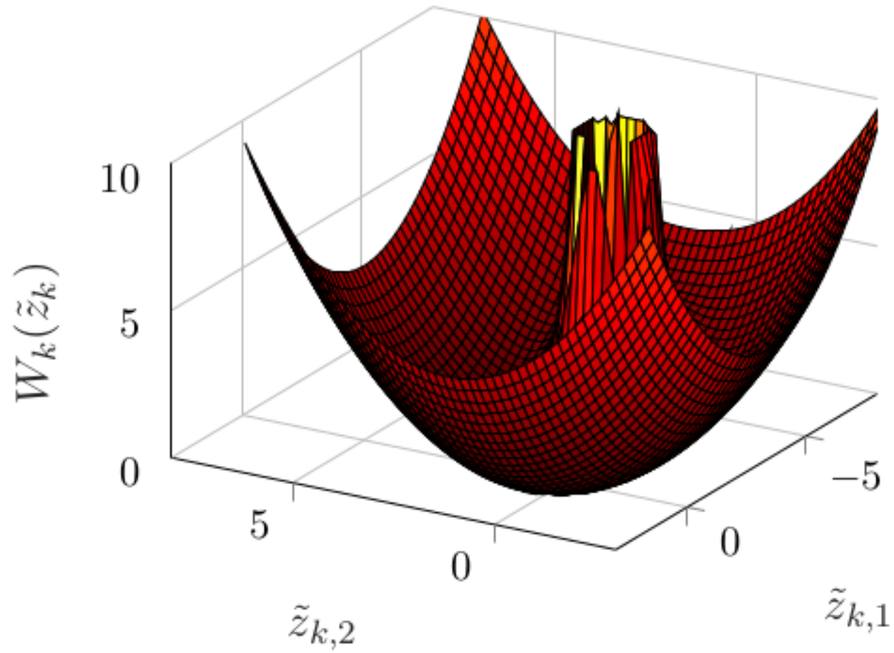
$\Delta_k = 50$

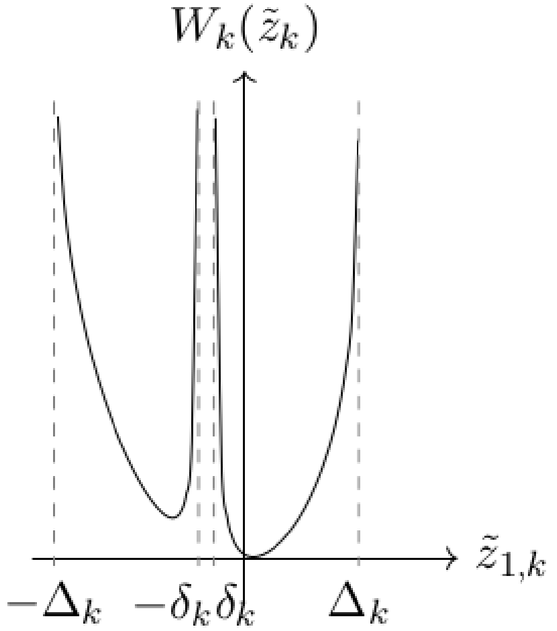


$\Delta_k = 20$



Implementation - Collision Avoidance

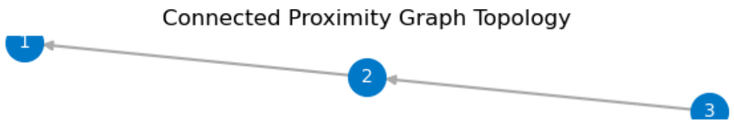




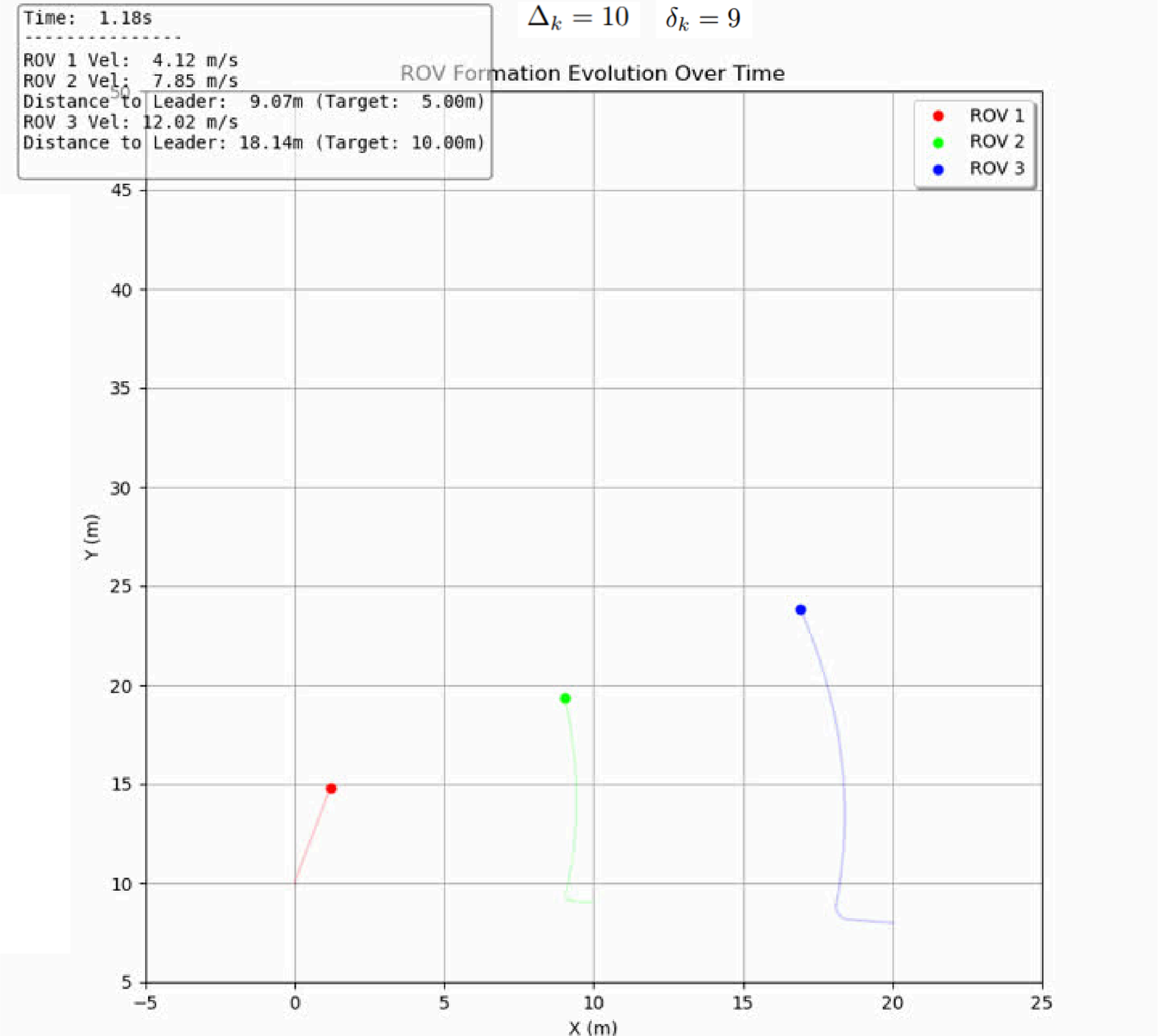
$$\nabla W(\tilde{z}) = \left[\left(\frac{1}{\Delta_k^2 - |z_k|^2}\right) - \left(\frac{1}{|z_k|^2 - \delta_k^2}\right)\right] z_k$$

Connectivity Maintenance

Collision Avoidance

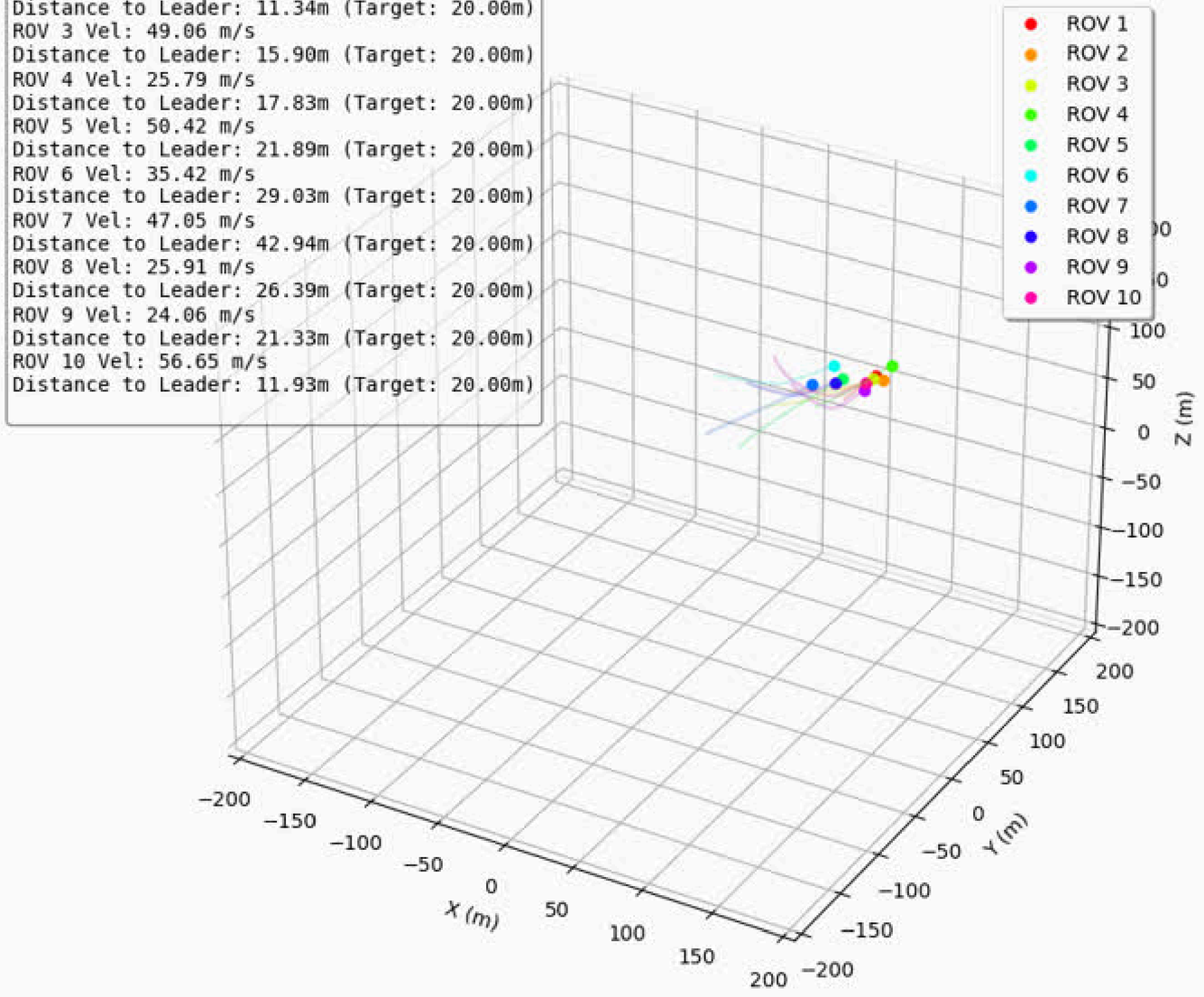


$\Delta_k = 10$ $\delta_k = 9$



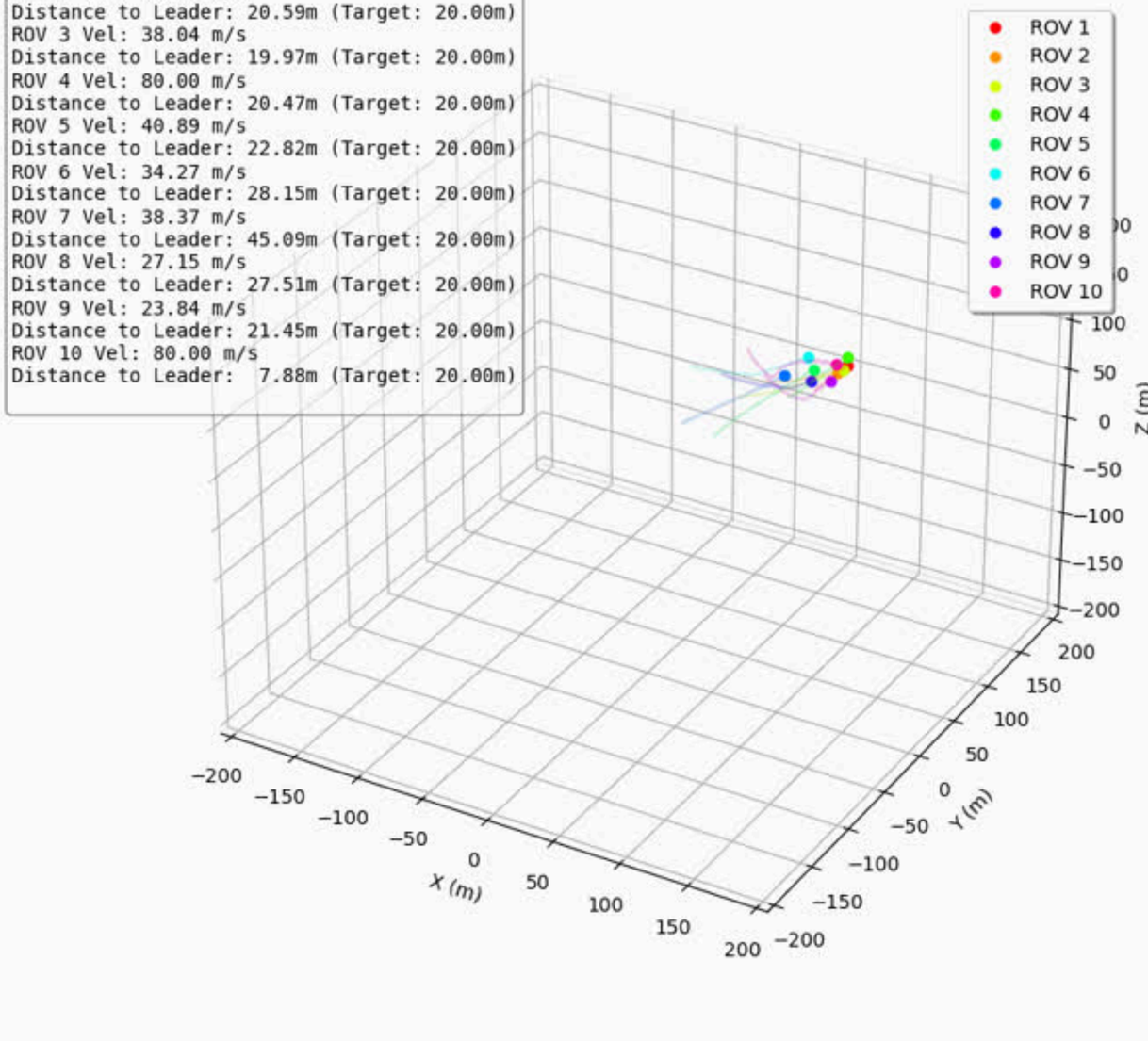
$\Delta_k = 50$ $\delta_k = 10$

```
Time:  1.42s
---------------
ROV 1 Vel: 21.21 m/s
ROV 2 Vel: 30.65 m/s
Distance to Leader: 11.34m (Target: 20.00m)
ROV 3 Vel: 49.06 m/s
Distance to Leader: 15.90m (Target: 20.00m)
ROV 4 Vel: 25.79 m/s
Distance to Leader: 17.83m (Target: 20.00m)
ROV 5 Vel: 50.42 m/s
Distance to Leader: 21.89m (Target: 20.00m)
ROV 6 Vel: 35.42 m/s
Distance to Leader: 29.03m (Target: 20.00m)
ROV 7 Vel: 47.05 m/s
Distance to Leader: 42.94m (Target: 20.00m)
ROV 8 Vel: 25.91 m/s
Distance to Leader: 26.39m (Target: 20.00m)
ROV 9 Vel: 24.06 m/s
Distance to Leader: 21.33m (Target: 20.00m)
ROV 10 Vel: 56.65 m/s
Distance to Leader: 11.93m (Target: 20.00m)
```



$\Delta_k = 50$ $\delta_k = 20$

```
Time:  1.50s
---------------
ROV 1 Vel: 21.21 m/s
ROV 2 Vel: 44.65 m/s
Distance to Leader: 20.59m (Target: 20.00m)
ROV 3 Vel: 38.04 m/s
Distance to Leader: 19.97m (Target: 20.00m)
ROV 4 Vel: 80.00 m/s
Distance to Leader: 20.47m (Target: 20.00m)
ROV 5 Vel: 40.89 m/s
Distance to Leader: 22.82m (Target: 20.00m)
ROV 6 Vel: 34.27 m/s
Distance to Leader: 28.15m (Target: 20.00m)
ROV 7 Vel: 38.37 m/s
Distance to Leader: 45.09m (Target: 20.00m)
ROV 8 Vel: 27.15 m/s
Distance to Leader: 27.51m (Target: 20.00m)
ROV 9 Vel: 23.84 m/s
Distance to Leader: 21.45m (Target: 20.00m)
ROV 10 Vel: 80.00 m/s
Distance to Leader:  7.88m (Target: 20.00m)
```



Next Steps

- Formation control with single integrator agents no constraints
  - Undirected graph
  - Directed graph
- Single integrator formation control with constraints
  - Only connectivity maintenance
  - Then also colision avoidance
- Double integrator agents
- BlueRov2 model no input constraints
- [Research challenge] integrate input constraints
- [Research challenge] controller robust or less sensitive to uncertainties (e.g. inverse optimal controller in the sense of LgV like control laws rather than feedback linearization)



Questions & Comments